Building a unified science of sensorimotor control with Densely Observable Measurment Environments [DOMEs]

FreeMoCap Foundation, Inc.

Written Proposal to the NSF X-Labs Initiative

The perception/action loop of biological and artificial agents follows the same pattern: sample a thin slice of the available environmental energy through a limited set of imperfect transducers, then generate reaction forces against a substrate in service of some goal. Information flows in, forces flow out; the brain exists to yank the bones around.

Myriad disciplines within the broad vehicle of science point their lenses at key components of this complex exchange sensation, perception, decision, and action. Perceptual neuroscientists built magnificently precise tools and theories to measuring and explicating the neural mechanisms behind the cascade of neural activity which follows the absorbtion of a photon by a opsin molecule in the retina. Musculoskeletal biomechanists have characterized the stretch/strain curves the winding `titan` filaments in our muscle spindles and mapped them to the full-body muscle activity of human movement [TODO - rewrite so its F&B reaching - photon - retina - V1 - etc, GRF - GTO - Muscle, motor - spine - muscle - force, or something].

Each member of these individual research clades understands that their work is contextualized in a broad and deeply cross-disciplinary domain of inquiry, and each strives to reach out across those disciplinary divides to their allies in nearby fields. But in practice, most of those threads fail to connect. Walk through the poster session of a visual neuroscience conference, you will mostly encounter variations of two-alternative-forced-choice-center-out-fixation tasks in non-mobile, head-fixed animal models. At the motor neuroscience conference, it will be single-joint planar reaching movements through a force field. At the biomechanics conference, its steady state walking on a treadmill. Each clade builds beautifully nuanced and complex theories in their domain of inquiry, but the pieces never quite fit together to form a coherent whole. Each group occupies a different room in the tower of Babel. They learned the nuances of different instruments, mastered different branches of mathematics. They are motivated by different kinds of arguments and ground their assertions in different flavors of statistics. They call out to each other in the first and last paragraphs of their research articles, the distant call of an unmet ally echoing across the yawning darkness between the fields.

And yet we are a species of explorers and we are drawn to those dark spaces. Each new generation of scientist builds new tools for their exploration; most dig down within their specialized niche, but some brave few build outwards towards the distant glowing shapes in the fog. Some of those even succeed, building new intergrations of previously disjointed theory - shedding light on new territory and the promise of new science at the intersections between deep domains of existing knowledge. But the discoverer cannot stay there. As reward for their success, they are bundled up and shipped to a new lab across the world with a rubber stamp on their forhead that reads "This One Knows Neuroscience." The tools their discovery - built through years of hard battles and tough lessons - are handed off the latest, least experienced member of the research group with the instruction "Pick up where they left off." And sometimes they do - bolting new complexity on an inherted pile of technical debt - until eventually it collapses under its own weight and is thrown onto an ever-growing pile of academic abandonware - unique and powerful tools that could have changed the face of science and human society… but never made it across the hall.

Mentorship-based academic research has it backwards - Tools should be built by masters and taugh tto students, not the other way around. We should bake the wisom of our experise and long experience into the tools put into the hands of students and other researchers.

Roboticists approach at a different scale, building swarms of flying drones comminucating and navigating in a delicate aerial dance and uncanny legged agents - after decades of work - are finally starting to take their first steps into real human environments.

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DRAFT - Work in Progress

1. Technology Landscape

- Existing tools and methods
  - Myriad single-modality hyperspecialized research threads
    - OpenSim - Widely used but universally hated by practictioners
    - DeepLabCut - Global success on paper, universally a headache in practice
    - etc
  - MuJoCo/IsaacLab Driving robotics
    - Integrate NeuroBehaioral Data into those sinulation/RL training enviornments to cross-align perceptuomotor neuroscience, HNHA biomechanics, robotics control research (e.g. Cassie from Guinea Fowl)
- New approaches to complexity management
  - Ontology/Entity Management (Anduril/Palantir/Maven Smart System)
  - Industry and Military technology applied to scientific domain
  - Want to establish an operational ontology as backbone of this tool, following powerful patterns established by (Anduril/Palantir/Maven) systems
- FreeMoCap and its landscape
  - Established Tool with Growing International Community
  - Existence is proof-of-concept of approach, XLabs scale support would accelerate FreeMoCap model into the global scale
- Laser Skeleton research
  - [1–3]
  - etc
  - Ferret work
    -

2. Outcomes

- Cool Tool - ~~Skinner~~SkellyBox: Multiscale modular empirical capture arena
  - Interoperabilty across species, domains, and timescales
    - A member of a mouse lab can walk into a marmoset lab and expect their pipelines will still work
    - Perceptuomotor behavior and kinematic/kinetic analysis directly interoperable with robotics models (via MuJoCo and Nvidia Isaac Gym) - Data for reinforcement learning models, Controls to interpret HNHA behavioral data
    - Data models alignment across timescales - developmental/longitudinal (months/years/lifespan) timescales aligned with perception/action timescales (seconds/minutes)
  - Composable complexity
    - Careful ontology allows for complexity via comopsible data sources and assocaited downstream entities
    - Adding new sensor modalities into established scene, framework handles spatiotemporal alignment/calibration, unit conversions, resampling, etc
- Organize and operationalize a new culture of science around discussion/management/extension shared platform tool
  - In the form of an annual Congress meeting
  - Organizing the culture of science around the discussion and advancement of shared ontologies rather than an endless stream of 8-10 page pdfs
  - Break the Skill Ceiling in academic science
    - Build tool to allow multi-decade continuous education across disciplines and experience (same tool in high school as used in federally funded research lab)
    - Shared toolset encourages natural blending of disciplines while encouraging and reward higher specialized exploration (esp when that research produces actionable results across different dimensional domains)
- Mesoscale revolution
  - Allow advances in reductionist microscale research to align with human-scale interests
  - Align Perceptuomotor Neuroscience with humanoid robotics, allow transfer, technology, and technique transfer between those fields

2.1. Timeline

Phase	Name	Description
1	**Organize**	Set up Organization (Phase 0) — Solidify release of FreeMoCap v2
2	**Architect**	Develop core software architecture, define key ontologies and entity/action pairings via use-oriented development (ferret, mouse, guinea fowl, marmoset, human research). Design and prototype instrumentation (mocap/eye cameras, electrode arrays and miniscopes, motor-unit EMG, environmental control systems)
3	**Execute**	Integrate new architecture into global/public facing community of users (presented as FreeMoCap v3). Iterative design of instrumented system
4	**Stabilize**	Establish OODA loop within expected standard operations
5	**Expand**	Once stabilized, expand key performance metrics
6	**Align**	Focus on building deeply integrated community of users and contributors (Transition BDFL ⟶ BDFaSPoT)
7	**Establish**	Focus on long term stability and healthy growth
8	**Maintain**	Beyond proposal timeline — Establish organization practice in service of unbounded infinite timeline. Apply winning model to support extending ‘mesoscale research revolution’ to all biological sciences (e.g. integrating deeper microbio/biochem insights beyond the scope of the present perceptuomotor/biomech/robotics focus)

Table 1: Timeline of Phased Milestones

3. Senior/Key Personnel Qualifications

3.1. Jonathan Matthis, PhD [Principle Investigator]

3.2. EI

3.3. AC,PhD

3.4. RR, CPA or w/e

Name / Role	Qualifications
Jonathan Matthis, PhD President/CEO	• Founder and chief maintainer of FreeMoCap project • Left tenure track academic research position precisely because the institution could not support the work that needed to be done • Decades experience integrating visual neuroscience with biomechanics/robotics[1–3]
EI CTO	• Decades of experience in software Industry • Chief architect and CTO for established companies • Represents expertise truly absent from academic contexts • Ensures world-class enterprise-level software capability • Will build training/workshop/hackathon backend to inject EI-level software development expertise into broad community of science
AC, PhD CSO	• Expertise in validation of clinical research tools and bench-to-bedside development • [CITE DISSERTATION] • Experience in neuroprosthetics design
RR CFO	• Bookkeeping • NSF Grant management • Finances • Entrepreneurship

Table 2: Senior/Key Personnel Qualifications

4. Team Capabilities Statement

4.1. Senior Team Capabilities

- Jon - done it before in humans, doing it in ferrets and mice
- AC - focusing clinical adoption and bench-to-bedside transitions
- EI - Deep professional experience in Tech Sector, high level software architecture experienced necessary to develop truly global scale enterprise level of software in open-source/academic contexts
- RR - Organizational health and entrepreneurial development

4.2. Collaborators

Initials	Model / Domain	Expertise
BS	Ferrets	Visual/Perceptual neuroscience, ephys
MD	Guinea Fowl	Musculoskeletal Biomechanics, muscle units
DF	Mice	System Biology
AH	Marmosets / NHP	Ephys
MH	Human	Vision and CPS
Mayo Clinic	Human	Brain scans and implanted sensors
BD, GN/RAI, CH, JH	Robots / Fabrication	Control theory
AA	Physical facilities	Rapid iteration, recruiting

Table 3: Collaborator Network

4.3. Governance

- Most successful FOSS project operate under a Benevolent Dictator For Life (BDFL) Governance (e.g. Linus Torvald at Linux)
- This is best when an organization is young, but later on centralization is a concern control should melt into the community of users (see Ton Roosenthaal's step down from Blender lead position)
- We will use Benevolent Dictator for a Set Period of Time (BDfaSPoT)
  - Transition from centralized control: President -> President + Board -> to President + Board (C-Suite) + Community DAO
    - President maintains at least 51% control until year 5, then optional transition to 50/50 President/Board+Community
- Plan changes via public Requests for Comments (similar to Python PEP system)
  - Provide public space for feedback on planned features, allows transparency without creating a bottleneck
- In 2nd year, create Developer Fund and Community Grants program
  - Stakeholders vote on proposals and features, via DAO like voting
- In 4-5th year, extend this system to handle community steering of org

4.4. Autonomy

- We are already autonomous, as of 2 weeks prior to the deadline of this proposal.
  - This project will happen no matter what, but with XLabs funding, we could truly change the world

Bibliography

1. Matthis, J. S., Yates, J. L. & Hayhoe, M. M. Gaze and the Control of Foot Placement When Walking in Natural Terrain. *Current Biology* **28,** 1224–1233.e5 (2018).
2. Matthis, J. S., Muller, K. S., Bonnen, K. L. & Hayhoe, M. M. Retinal Optic Flow during Natural Locomotion. *PLOS Computational Biology* **18,** e1009575 (2022).
3. Muller, K. S. *et al.* Retinal Motion Statistics during Natural Locomotion. *eLife* **12,** e82410 (2023).